and much of his reasoning had been carried out in verbal give-and-take with Baade, Zwicky and Baade agreed to present their work together at a meeting of the American Physical Society at Stanford University, an easy day's drive up the coast from Pasadena. The abstract of their talk, published in the 15 January 1934 issue of the *Physical Review*, is shown in Figure 5.2. It is one of the most prescient documents in the history of physics and astronomy.

It asserts unequivocally the existence of supernovae as a distinct class of astronomical objects—although adequate data to prove firmly that they are different from ordinary novae would be produced by Baade and Zwicky only four years later, in 1938. It introduces for the first time the name "supernovae" for these objects. It estimates, correctly, the total energy released in a supernova. It suggests that cosmic rays are produced by supernovae—a hypothesis still thought plausible in 1993, but not firmly established (see Footnote 3). It invents the concept of a star made out of neutrons—a concept that would not become widely accepted as theoretically viable until 1939 and would not be verified observationally until 1968. It coins the name *neutron star* for this concept. And it suggests "with all reserve" (a phrase presumably inserted by the cautious Baade) that supernovae are produced by the transformation of ordinary stars into neutron stars—a suggestion that would be shown theoretically viable only in the early 1960s and would be confirmed by observation only in the late 1960s with the discovery of *pulsars* (spinning, magnetized neutron stars) inside the exploding gas of ancient supernovae.